

# SMI3.0 Data Quality, Suitability and Preprocessing Validation Report

Audit-ready assessment based on the provided validation template, policy processor, borrower processor, merge processor, DataManager, data\_utils, and preprocess code.

Prepared date: July 8, 2026

## Executive Summary

The reviewed codebase implements a structured mortgage-insurance data preparation pipeline covering source loading, MI/LRU Plus population construction, borrower and credit-score aggregation, preprocessing, economic feature integration, target labeling, feature engineering, and final dataset output. The assessment concludes that the data are fit for model development and validation subject to remediation of documentation, reconciliation, leakage-control, and reproducibility gaps.

- Overall validation conclusion: Fit for use subject to conditions.
- Most code-level controls are implemented for filtering, merging, preprocessing, outlier capping, target creation, and report checks.
- Key gaps: no formal record-count reconciliation from source to target, no extraction-date evidence, incomplete run-log evidence, limited independent verification of labels, and potential forward-looking economics features that must be explicitly excluded from training or reserved for backtesting/stress analysis.
- Highest-risk technical issues: inconsistent notebook versus refactored pipeline naming/logic, potential leakage from HPI\_+ and UE\_+ features, use of post-event/current portfolio outcomes in target labeling without a documented observation/performance window, and reliance on notebooks with embedded outputs rather than a fully controlled production run artifact.

| Template Section                     | Assessment            | Summary                                                                                                                                       |
|--------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose and Scope                    | Pass                  | Scope is consistent with the template and codebase coverage.                                                                                  |
| Data Sources and Lineage             | Partial               | Sources and joins are identifiable, but extraction dates, source-to-target mapping, and reconciliation evidence are incomplete.               |
| Data Completeness Review             | Partial               | Some missingness is quantified and treated, but no complete variable-level completeness inventory is provided.                                |
| Data Accuracy and Consistency Review | Partial               | Logical cleaning, SID repair, clipping, and duplicates are addressed; independent reconciliation is incomplete.                               |
| Data Relevance and Suitability       | Partial               | Portfolio, product, geography, vintage, and status representation are documented, but bias and representativeness tests are limited.          |
| Target Label Construction            | Partial               | Target logic is implemented; observation/performance windows and leakage controls need stronger evidence.                                     |
| Preprocessing Review                 | Partial               | Pipeline is reproducible in code, but not fully traceable to locked parameters and run logs.                                                  |
| Missing Value Treatment              | Partial               | Several treatments are justified; broad missingness impact analysis is incomplete.                                                            |
| Outlier Treatment                    | Partial               | Capping is configurable and breach counts exist; business approval and performance impact evidence are incomplete.                            |
| Business Rules and Aggregation       | Partial               | Borrower aggregation and product rules are implemented; some manual/business overrides require formal documentation.                          |
| Bias and Data Leakage Assessment     | Partial / Conditional | Known selection/reject-inference and future-feature risks require remediation before reliance.                                                |
| Reproducibility                      | Partial               | Version notes and output files exist, but formal version control, environment capture, and approved-version reconciliation are not evidenced. |
| Conclusion                           | Conditional Pass      | Data fit for use subject to remediation and control conditions.                                                                               |

## 1. Purpose and Scope

| Review Item                                                | Status  |
|------------------------------------------------------------|---------|
| The model purpose has been clearly documented              | Partial |
| The target population has been clearly defined             | Partial |
| The development period is appropriate for the intended use | Partial |

|                                                                 |     |
|-----------------------------------------------------------------|-----|
| The data sources have been identified and documented            | Yes |
| The scope of the review is appropriate for the model complexity | Yes |

#### Completed validation comments:

The framework evaluates the mortgage insurance portfolio data used by SMI3.0, including policies, borrowers, credit bureau scores, economic variables, and LRU Plus. The code supports an end-to-end build sequence from source loading to preprocessing, merging, target labeling, feature engineering, and duplicate handling. However, the model purpose, target population, development period, and intended model-use boundaries should be explicitly stated in the validation package, not inferred from code and notebook comments.

## 2. Data Sources and Lineage

| Review Item                                       | Status   |
|---------------------------------------------------|----------|
| All data source systems have been identified      | Yes      |
| Data extraction dates have been documented        | No / Gap |
| Source-to-target mapping has been provided        | Partial  |
| Record count reconciliation has been completed    | Partial  |
| Filters, joins, and exclusions are documented     | Partial  |
| Manual interventions are documented and justified | Partial  |
| Final dataset can be reproduced                   | Partial  |

#### Completed validation comments:

The code identifies policy, borrower, TransUnion/FICO, CreditVision, economics, Toronto FSA, ETD claim/delinquency, and LRU Plus files. DataManager centralizes parquet/CSV reads, derives received/effective SIDs, filters MI policies, builds LRU records, merges borrower, TT, and economics data, applies windows, labels targets, and drops duplicate applications. Notebooks provide some shape checks and report logs, including final combined dataset rows, years, regions, LOB, and status. Gaps remain because extraction timestamps, formal source-to-target mapping, and row-count reconciliation after each filter/join are not consistently captured.

## 3. Data Completeness Review

| Review Item                                           | Status  |
|-------------------------------------------------------|---------|
| Missingness has been quantified for key variables     | Partial |
| Missingness reviewed by segment and geography         | Partial |
| Missingness is within acceptable tolerance            | Partial |
| Missingness patterns align with business expectations | Partial |
| No major concentration of missing values identified   | Partial |
| Elevated missingness assessed for impact              | Partial |

#### Completed validation comments:

Completeness is partly assessed. The merge notebook quantifies high missingness for PRK\_CONDO\_FEE and CRA\_UNSATISFIED\_COLLECTION\_AMT and explains why zero-fill is business-meaningful. Report logs summarize geography and portfolio coverage. However, the evidence does not include a full key-variable missingness table, missingness by vintage/region/product/status, tolerance thresholds, or a measured model-impact assessment for elevated missingness.

## 4. Data Accuracy and Consistency Review

| Review Item                                     | Status        |
|-------------------------------------------------|---------------|
| Key variables tested for logical consistency    | Partial       |
| Date sequencing is valid                        | Partial       |
| Derived fields reconcile to source fields       | Partial       |
| Duplicate records identified and addressed      | Yes           |
| Impossible values handled appropriately         | Yes / Partial |
| Field definitions are consistent across systems | Partial       |

#### Completed validation comments:

The code handles common accuracy issues: converts IDs and dates, repairs SID zero pairs, filters out invalid policy status groups, coerces bureau scores and numeric fields, clips invalid credit scores, normalizes rates, caps configured feature ranges, handles impossible amortization and term values, derives property age/value, and drops duplicate applications. The ETD file is explicitly collapsed to one row per application. Remaining gaps include no independent data dictionary mapping, limited derived-field reconciliation evidence, and inconsistent variable names between notebooks and refactored modules.

## 5. Data Relevance and Suitability

| Review Item                                     | Status        |
|-------------------------------------------------|---------------|
| Development sample reflects intended population | Partial       |
| Product types are appropriately represented     | Partial       |
| Geographic coverage is appropriate              | Yes / Partial |
| Vintage coverage is sufficient                  | Yes / Partial |
| Stress and benign periods represented           | Partial       |
| No major survivorship bias identified           | Partial       |

#### Completed validation comments:

The final combined report check shows broad geography coverage, MI/LRU Plus segmentation, funded/declined statuses, and years 2009-2025. Product-group rules identify regular, ALTA, HELOC, borrowed-down, cashback, vacation/second, new-to-Canada, credit-assist, and HB100 groups, with ALTA dropped in the merge processor. Relevance is generally supportable, but suitability remains conditional because product exclusions, good-decline filtering, LRU-specific inclusion, reject inference, and stress/benign period representation require formal business sign-off and quantified distribution review.

## 6. Target Label Construction

| Review Item                                   | Status         |
|-----------------------------------------------|----------------|
| Target variable definition documented         | Yes / Partial  |
| Observation and performance windows specified | Partial        |
| Event timing is appropriate                   | Partial        |
| No future information used in labels          | Partial / Risk |
| Outcome assignment independently verified     | No / Gap       |

#### Completed validation comments:

The target is defined as any non-No\_Event category, where declined policies, claims, and 90-day delinquency records become target=1. ETD episodes are filtered using BOOKYEAR-specific date rules and collapsed by application using max claim/delinquency flags and summed incurred loss. This is reasonable, but leakage risk remains unless the observation date, performance window, and label availability rules are formally documented and tested. Declines are included as an event class, which may be appropriate for risk ranking but should be justified because declines can reflect underwriting decisions rather than future borrower performance.

## 7. Preprocessing Review

| Review Item                                         | Status        |
|-----------------------------------------------------|---------------|
| Preprocessing steps documented                      | Yes / Partial |
| Filtering rules justified                           | Partial       |
| Deduplication logic appropriate                     | Yes           |
| Transformations suitable for variable distributions | Partial       |
| Binning and grouping reasonable                     | Partial       |
| Scaling and normalization appropriate               | Partial       |
| No major preprocessing bias identified              | Partial       |
| Preprocessing pipeline reproducible                 | Partial       |

#### Completed validation comments:

The Preprocessor applies an ordered set of vectorized steps: tenure mapping, property age/value computation, rural assignment, economic-region mapping, amortization/term repair, rate normalization, configurable range clipping, loan-purpose/refinance normalization, portability normalization, occupancy normalization, and insurance-in-force recomputation. The design is structured and reproducible in code. Controls remain partial because several default steps are commented out, business rationale for each transformation is not consistently documented, and notebook logic is not fully aligned with the refactored pipeline.

## 8. Missing Value Treatment

| Review Item                               | Status         |
|-------------------------------------------|----------------|
| Missing values treated consistently       | Partial        |
| Imputation rationale documented           | Partial        |
| Missingness indicators considered         | No / Gap       |
| Dropped observations justified            | Partial        |
| No future information used for imputation | Partial / Risk |
| Impact of missing treatment assessed      | Partial        |

#### Completed validation comments:

Missing treatment includes replacing sentinel bureau values with NaN, filling claim/delinquency indicators with zero when no ETD match exists, filling selected features with zero where missing has business meaning, imputing contract/qualifying rates from rate-type/year group means, imputing property age from valid existing-property averages, and filling credit scores through hierarchical bureau fallbacks. These are defensible but must be documented as model assumptions. Missingness indicators are not created for major imputations, and group-mean imputation should be confirmed to use only development-period information available at model scoring time.

## 9. Outlier Treatment

| Review Item                            | Status        |
|----------------------------------------|---------------|
| Outlier thresholds documented          | Yes / Partial |
| Consistent methodology used            | Yes / Partial |
| Extreme values reviewed for validity   | Partial       |
| Capping methods justified              | Partial       |
| Observation removal justified          | Partial       |
| Impact on performance reviewed         | No / Gap      |
| Valid tail-risk observations preserved | Partial       |

#### Completed validation comments:

Outlier handling is implemented through configured capping ranges and bureau-score validity ranges. The merge processor produces breach counts and breach percentages for variables such as GDS, TDSR, collection amounts, ACTLTV, income, loan term, property value, and contract rate. This is strong evidence of systematic treatment. The main gaps are lack of formal approval for thresholds, lack of before/after distribution plots, no sensitivity analysis, and potential concern that capping tail-risk credit variables may attenuate valid stress signals.

## 10. Business Rules and Aggregation

| Review Item                           | Status        |
|---------------------------------------|---------------|
| Aggregation rules documented          | Yes / Partial |
| Record consolidation logic is correct | Partial       |
| Exposure duplication ruled out        | Partial       |
| Date alignment logic appropriate      | Partial       |
| Loan-level mapping accurate           | Partial       |
| Manual overrides documented           | Partial       |

|                                        |         |
|----------------------------------------|---------|
| Aggregation preserves economic meaning | Partial |
|----------------------------------------|---------|

#### Completed validation comments:

Borrower aggregation groups by application number, creates NUM\_OF\_BOR, clips/falls back credit scores, and produces min/max/avg/weighted-average statistics. ETD episodes are consolidated to one application row. MI/LRU construction, product grouping, economics merging, TT effect SID override, and duplicate removal are implemented. However, validation should add join-cardinality checks, duplicate exposure checks before/after merge, and formal rationale for the use of keep=last, TT overrides, LRU inclusion rules, and product-group exclusions.

## 11. Bias and Data Leakage Assessment

| Review Item                              | Status         |
|------------------------------------------|----------------|
| Sampling bias assessed                   | Partial        |
| Selection bias assessed                  | Partial        |
| Survivorship bias assessed               | Partial        |
| Reject inference considered              | Partial / Risk |
| Exclusions documented and justified      | Partial        |
| Segment-level differences reviewed       | Partial        |
| Identified bias assessed for materiality | No / Gap       |

#### Completed validation comments:

The pipeline includes declined applications, funded applications, MI, and LRU Plus, and report logs summarize year/region/status/LOB composition. This helps identify broad segment coverage. Material bias risks remain: good-decline filtering, ALTA/product exclusions, LRU selection, inclusion of declines as target events, and potential geographic/product/vintage imbalance. Leakage risk is material where HPI\_+ and UE\_+ features are generated; they must be explicitly excluded from training inputs unless used only for backtesting/stress/labeling analysis. Feature lists used in model training should be frozen and checked for future-looking variables.

## 12. Reproducibility

| Review Item                           | Status        |
|---------------------------------------|---------------|
| Code or logic version controlled      | Partial       |
| Parameters documented                 | Partial       |
| Input and output files identified     | Yes / Partial |
| Run logs available                    | Partial       |
| Final dataset can be recreated        | Partial       |
| Results reconcile to approved version | No / Gap      |

#### Completed validation comments:

The notebooks contain version histories, configuration-driven inputs, output filenames, and report\_log summaries. DataManager is a refactored code path with centralized configuration and save/load behavior. However, there is no evidence of git commit IDs, environment lockfiles, immutable input snapshots, extraction dates, automated run logs, unit tests, or approved-version reconciliation. Reproducibility is therefore code-supported but not yet audit-complete.

## 13. Conclusion

| Review Item                                | Status                       |
|--------------------------------------------|------------------------------|
| No material issues identified              | No                           |
| Issues are not material to fit-for-use     | Conditional                  |
| Issues require remediation before reliance | Yes, for production reliance |
| Data are fit for use                       | Conditional                  |
| Data are fit for use subject to conditions | Yes                          |

|                                       |          |
|---------------------------------------|----------|
| Results reconcile to approved version | No / Gap |
|---------------------------------------|----------|

### Completed validation comments:

The data preparation framework is directionally sound and sufficiently mature for controlled model development and validation analysis. The dataset should be considered fit for use subject to conditions: complete lineage and reconciliation evidence, confirm no forward-looking features enter training, document target windows and decline treatment, finalize missing/outlier treatment rationale, and establish reproducible run controls with approved-version reconciliation.

## Code Review Findings and Risks

| Severity | Finding                                                          | Validation Impact / Required Action                                                                                                                                                                                                         |
|----------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High     | Potential leakage from future economic variables                 | The economics preparation creates UE_+s and HPI_+s fields using forward shifts. These should be removed from model training features or limited to backtesting/stress/label construction with explicit controls.                            |
| High     | Target definition requires formal observation/performance window | Decline, claim, and delinquency outcomes are combined into target=1. This must be supported by a documented window and verified event timing.                                                                                               |
| High     | Notebook and refactored code divergence                          | The notebooks reference PRD_QUAL_INT_RATE, RURAL_IND, CLAIM_IND/DELQ90_IND, and several product rules, while DataManager/preprocess use PRD_CONTRACT_RATE, RURAL, claim_row/delq_row, and a different code path. A parity test is required. |
| Medium   | Record-count reconciliation incomplete                           | Some shapes are printed, but a formal reconciliation table covering source, filter, join, exclusion, duplicate removal, and final output is missing.                                                                                        |
| Medium   | Outlier capping thresholds need governance approval              | Thresholds are applied and breach counts exist, but business justification, sensitivity testing, and tail-risk preservation evidence are incomplete.                                                                                        |
| Medium   | Missing-value treatment lacks global inventory                   | Selected high-missing variables are treated, but a full missingness matrix and missingness-by-segment assessment are not documented.                                                                                                        |
| Medium   | Credit-score fallback hierarchy should be validated              | Borrower aggregation propagates group values and falls back to CV and TU scores. Validate timing, source priority, and whether fallback can introduce stale/future bureau data.                                                             |
| Medium   | Good-decline and ALTA exclusion rules may create selection bias  | Excluded populations should be profiled, justified, and monitored for representativeness impact.                                                                                                                                            |
| Low      | Use of notebooks as primary artifacts                            | Notebooks contain embedded outputs and manual execution state. Convert to deterministic scripts or pipelines with unit tests and run manifests.                                                                                             |
| Low      | Run logs and environment capture incomplete                      | Add run ID, input file hashes, code commit, package versions, config hash, and output file checksum to every build.                                                                                                                         |

## Remediation Plan

| Priority | Recommended Remediation                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Create an end-to-end lineage matrix mapping each final field to source table/file, join key, transformation, and owner.                                         |
| 2        | Add automated row-count reconciliation after every load, filter, join, aggregation, deduplication, and write step.                                              |
| 3        | Freeze the model feature list and add an automated leakage test blocking fields with + horizons or known outcome/post-event fields.                             |
| 4        | Document and independently test target construction, including observation date, performance window, event-date cutoff, decline treatment, and ETD aggregation. |
| 5        | Produce missingness tables by key variable, product group, LOB, geography, vintage, status, and target.                                                         |
| 6        | Add threshold governance for capping, including before/after plots, breach counts, business sign-off, and model sensitivity testing.                            |
| 7        | Add unit tests and parity tests between notebooks and DataManager/preprocess outputs.                                                                           |
| 8        | Capture reproducibility metadata: code commit, config hash, input file names/hashes, extraction dates, package versions, run timestamp, and output checksum.    |
| 9        | Perform bias/reject-inference review for declined applications, product exclusions, LRU Plus, geography, and vintage distributions.                             |
| 10       | Reconcile final dataset to an approved version and retain the reconciliation evidence with the validation package.                                              |

## Evidence Appendix

| Evidence Area              | Source Evidence Used                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------|
| Template purpose and scope | Template states the review covers completeness, accuracy, relevance, representativeness, and |

|                                         |                                                                                                                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | preparation of MI portfolio data including policies, borrowers, credit bureau, economic, and LRU Plus data. Source lines: template L997-L1003.                                                                                            |
| DataManager build order                 | DataManager loads sources, builds MI/LRU, applies preprocessing, merges borrower/TT/economics, applies window, labels targets, engineers features, and drops duplicate applications. Source lines: data_manager L73-L91.                  |
| DataManager source loading              | DataManager loads policy, TT, CV scores, LRU, economics, borrower, FICO10/credit score, and Toronto FSA data. Source lines: data_manager L106-L195.                                                                                       |
| Preprocessor steps                      | Preprocessor default steps include tenure, property age/value, rural, econ region, amortization/term, rate normalization, range clipping, loan purpose/refi, port, occupancy, and in-force recompute. Source lines: preprocess L750-L770. |
| Target labeling                         | DataManager target labels declined, delq/claim, claim only, delq only, and no event. Source lines: data_manager L385-L420.                                                                                                                |
| Merge notebook report log               | Final combined dataset report contains rows, target rate, years, regions, LOB, and status. Source lines: merge notebook L355-L379.                                                                                                        |
| Outlier breach counts                   | Merge notebook includes breach counts and percentages by capped feature. Source lines: merge notebook L93-L116.                                                                                                                           |
| Borrower aggregation                    | Borrower processor documents source domains, credit-score cleaning, merge, borrower counts, score clipping, binning, and aggregation. Source lines: borrower notebook L831-L863 and L1543-L1612.                                          |
| Policy processor versioning and sources | Policy processor contains version history and documents policy, LRU+, and Toronto FSA data sources. Source lines: policy notebook L1643-L1672 and L1744-L1758.                                                                            |